

## MAYO, YT, Canada

WMO: 719650

Lat: 63.616N

Lon: 135.880W

Elev: 504

StdP: 95.42

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -47.4      | -43.9 | -43.3                         | 0.1 | -39.1 | -39.8 | 0.1 | -36.2 | 7.7                   | -14.2 | 6.9 | -14.1 | 0.1                   | 30   | 0.670 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.9                   | 27.1            | 15.2 | 25.1 | 14.3 | 23.2 | 13.6 | 16.2                | 24.5 | 15.3 | 22.6 | 14.5 | 21.1 | 2.3                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.6                            | 10.3 | 17.0 | 12.6 | 9.6 | 16.4 | 11.7 | 9.1 | 16.0 | 47.3          | 24.7 | 44.5 | 22.8 | 42.3 | 21.2 | 20.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.2           | 6.2  | 5.4 | DB                         | -47.9 | 29.6               | 3.5 | 2.0                                                | -50.5 | 31.0       | -52.6 | 32.2       | -54.5 | 33.4       | -57.1 | 34.8 | (4) |
| (5)               |      |     | WB                         | -41.0 | 18.0               | 2.9 | 1.2                                                | -43.1 | 18.9       | -44.8 | 19.6       | -46.4 | 20.3       | -48.5 | 21.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -2.0        | -22.5 | -17.5 | -10.0 | 1.2   | 9.1   | 14.7  | 16.0  | 13.2  | 7.0   | -2.1  | -15.3 | -19.4 | (6)  |  |
|                                                                        | DBStd       | 15.55       | 12.25 | 10.00 | 8.61  | 5.18  | 4.33  | 3.23  | 2.49  | 3.36  | 3.78  | 5.58  | 9.45  | 10.81 | (7)  |  |
|                                                                        | HDD10.0     | 4890        | 1006  | 770   | 621   | 264   | 70    | 2     | 1     | 7     | 102   | 376   | 758   | 913   | (8)  |  |
|                                                                        | HDD18.3     | 7458        | 1264  | 1003  | 879   | 513   | 287   | 117   | 79    | 163   | 340   | 633   | 1008  | 1171  | (9)  |  |
|                                                                        | CDD10.0     | 494         | 0     | 0     | 0     | 1     | 42    | 143   | 187   | 108   | 12    | 1     | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 21          | 0     | 0     | 0     | 0     | 1     | 8     | 7     | 4     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 397         | 0     | 0     | 0     | 0     | 27    | 131   | 148   | 89    | 1     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 73          | 0     | 0     | 0     | 0     | 3     | 27    | 22    | 21    | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 1.9         | 1.4   | 1.7   | 2.6   | 2.6   | 2.3   | 1.9   | 1.5   | 1.5   | 1.8   | 2.1   | 1.7   | 1.6   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 328         | 18    | 15    | 10    | 10    | 24    | 41    | 55    | 44    | 37    | 30    | 22    | 21    | (15) |  |
|                                                                        | PrecMax     | 448         | 41    | 42    | 34    | 33    | 46    | 86    | 111   | 105   | 82    | 68    | 40    | 46    | (16) |  |
|                                                                        | PrecMin     | 149         | 2     | 2     | 1     | 0     | 4     | 11    | 13    | 12    | 12    | 7     | 3     | 6     | (17) |  |
|                                                                        | PrecStd     | 60          | 10    | 10    | 8     | 8     | 11    | 17    | 24    | 21    | 17    | 14    | 8     | 10    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 4.4   | 4.7   | 9.9   | 17.5  | 26.5  | 29.9  | 29.2  | 29.2  | 20.2  | 12.9  | 6.5   | 3.9   | (19) |  |
|                                                                        |             | MCWB        | 1.5   | 1.1   | 4.4   | 8.8   | 13.5  | 16.7  | 16.0  | 16.8  | 12.0  | 8.0   | 2.5   | 1.2   | (20) |  |
|                                                                        | 2%          | DB          | 0.3   | 1.3   | 6.3   | 14.0  | 23.0  | 26.7  | 26.7  | 25.9  | 17.3  | 8.8   | 1.3   | 0.3   | (21) |  |
|                                                                        |             | MCWB        | -1.0  | -0.5  | 1.7   | 6.4   | 11.9  | 14.8  | 15.0  | 15.5  | 10.5  | 5.0   | -0.1  | -0.9  | (22) |  |
|                                                                        | 5%          | DB          | -3.0  | -1.1  | 4.1   | 11.5  | 20.4  | 24.5  | 25.0  | 22.8  | 15.5  | 6.6   | -1.1  | -2.1  | (23) |  |
|                                                                        |             | MCWB        | -3.9  | -2.5  | 0.6   | 4.8   | 10.6  | 13.5  | 14.7  | 14.1  | 9.6   | 3.9   | -2.0  | -3.2  | (24) |  |
|                                                                        | 10%         | DB          | -6.7  | -3.7  | 1.7   | 9.4   | 18.0  | 22.4  | 23.1  | 20.4  | 13.8  | 4.7   | -3.3  | -5.2  | (25) |  |
|                                                                        |             | MCWB        | -7.3  | -4.9  | -0.9  | 3.5   | 9.4   | 12.7  | 14.1  | 13.2  | 8.8   | 2.6   | -4.1  | -5.9  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 2.0   | 1.4   | 4.5   | 9.0   | 14.2  | 17.3  | 17.5  | 17.5  | 13.1  | 8.2   | 2.9   | 1.5   | (27) |  |
|                                                                        |             | MCDB        | 4.1   | 3.9   | 9.2   | 17.0  | 24.6  | 27.6  | 25.5  | 27.4  | 18.0  | 12.3  | 5.5   | 3.3   | (28) |  |
|                                                                        | 2%          | WB          | -0.7  | -0.4  | 2.0   | 6.7   | 12.4  | 15.7  | 16.1  | 16.1  | 11.5  | 5.5   | 0.1   | -0.6  | (29) |  |
|                                                                        |             | MCDB        | 0.8   | 1.4   | 5.4   | 13.3  | 21.5  | 24.4  | 23.4  | 24.6  | 15.9  | 8.0   | 1.4   | 0.8   | (30) |  |
|                                                                        | 5%          | WB          | -3.3  | -2.3  | 0.7   | 5.1   | 11.1  | 14.5  | 15.4  | 14.9  | 10.2  | 4.1   | -1.9  | -2.9  | (31) |  |
|                                                                        |             | MCDB        | -2.3  | -1.0  | 4.0   | 10.8  | 19.1  | 21.8  | 22.4  | 21.4  | 14.4  | 6.2   | -1.0  | -1.8  | (32) |  |
|                                                                        | 10%         | WB          | -6.5  | -4.7  | -0.7  | 3.8   | 9.9   | 13.5  | 14.7  | 13.8  | 9.3   | 2.7   | -4.0  | -5.5  | (33) |  |
|                                                                        |             | MCDB        | -5.9  | -3.4  | 1.8   | 8.9   | 17.1  | 20.5  | 21.4  | 18.9  | 13.0  | 4.5   | -3.3  | -4.7  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.9   | 10.9  | 12.7  | 11.5  | 12.4  | 12.5  | 11.9  | 11.5  | 10.3  | 7.0   | 7.7   | 9.0   | (35) |  |
|                                                                        |             | MCDBR       | 10.5  | 10.1  | 11.4  | 13.7  | 15.6  | 15.7  | 16.0  | 16.0  | 13.1  | 9.1   | 8.5   | 11.2  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 9.5   | 8.7   | 7.9   | 7.8   | 7.3   | 6.7   | 6.8   | 7.3   | 7.4   | 6.1   | 7.6   | 10.2  | (37) |  |
|                                                                        |             | MCWBR       | 9.9   | 9.9   | 11.0  | 12.8  | 14.3  | 13.6  | 13.5  | 14.1  | 11.3  | 8.2   | 8.1   | 11.1  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.195 | 0.240 | 0.258 | 0.337 | 0.367 | 0.376 | 0.384 | 0.383 | 0.308 | 0.264 | 0.186 | 0.173 | (40) |  |
|                                                                        |             | taud        | 1.973 | 2.237 | 2.269 | 2.099 | 2.299 | 2.343 | 2.334 | 2.340 | 2.540 | 2.432 | 1.984 | 1.923 | (41) |  |
|                                                                        | Edn at Noon |             | 417   | 718   | 854   | 830   | 842   | 842   | 821   | 784   | 799   | 690   | 419   | 213   | (42) |  |
|                                                                        |             | Edh at Noon | 29    | 56    | 82    | 123   | 112   | 110   | 108   | 98    | 65    | 46    | 27    | 13    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.22  | 1.00  | 2.76  | 4.75  | 5.41  | 5.62  | 4.88  | 3.84  | 2.39  | 0.99  | 0.29  | 0.09  | (44) |  |
|                                                                        | RadStd      |             | 0.01  | 0.07  | 0.16  | 0.25  | 0.27  | 0.34  | 0.33  | 0.39  | 0.25  | 0.14  | 0.04  | 0.00  | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (2 neighbors) | N/A   | +2.16   | -3.09  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page